

Making Inferences about Features

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Q&A

| Logistics impact of moving online?

- We'll continue normal class meetings. **Your job** is to ask questions, interrupting if necessary.

| Projects?

- Our **final exam slot** is Friday Dec. 11 at 1:30 pm.
- Final project presentations will be virtual, details TBA

Q&A

| Is web scraping legal??

- Publicly available data? **Yes.** (see **EFF article**)
- If you need a login? Maybe. Ask for permission.

| How hard is it to scrape Twitter?

Constraints about what you can access without paying. But **tools available**

| What's a good Python dev environment?

I use: RStudio when integrating with R, VS Code for command-line or web dev, Jupyter Notebooks for pure-Python data science.

What's your goal?

- **Predict** unseen labels
 - How much will this house sell for?
 - Does this child have autism?
 - Is this a positive or negative movie review?
- **Infer** relationships between features and labels
 - How much does home size affect price?
 - Is DNA methylation a marker of autism?
 - Does "sick" indicate a positive or negative review?
- Understand the **causal** effect of interventions
 - How much will building an addition increase the price of my home?
 - Will antioxidants prevent autism?
 - Will cutting this scene make my movie get better reviews?

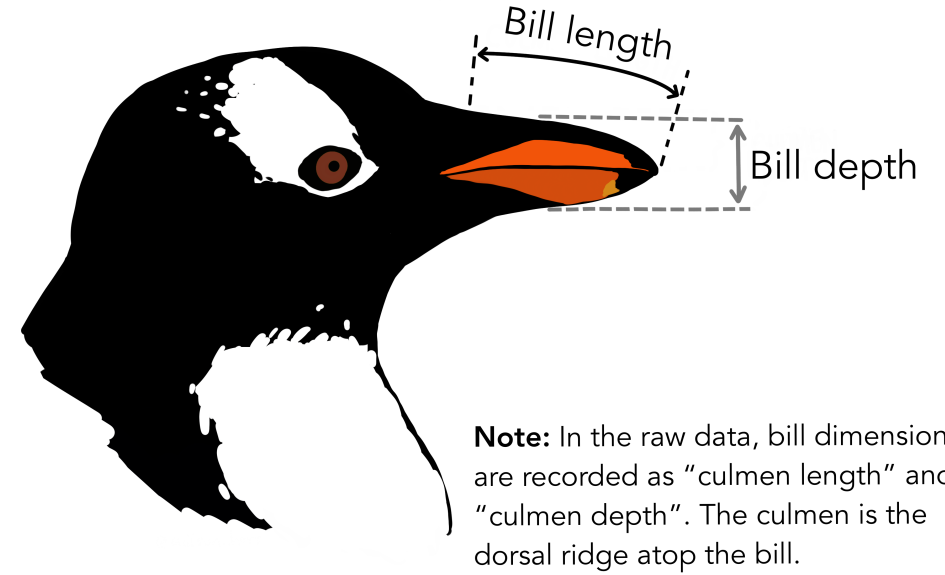
Techniques for Inference

- Classical statistical inference
 - 2-sample t tests, chi squared tests, ANOVA, ...
 - inference about model parameters (coefficient standard errors etc.)
- Variable importance plots
- Benefit of adding each feature

Objectives for Today

- Identify several different approaches for drawing conclusions about features
- Recognize potential challenges in making those inferences

Palmer Penguins



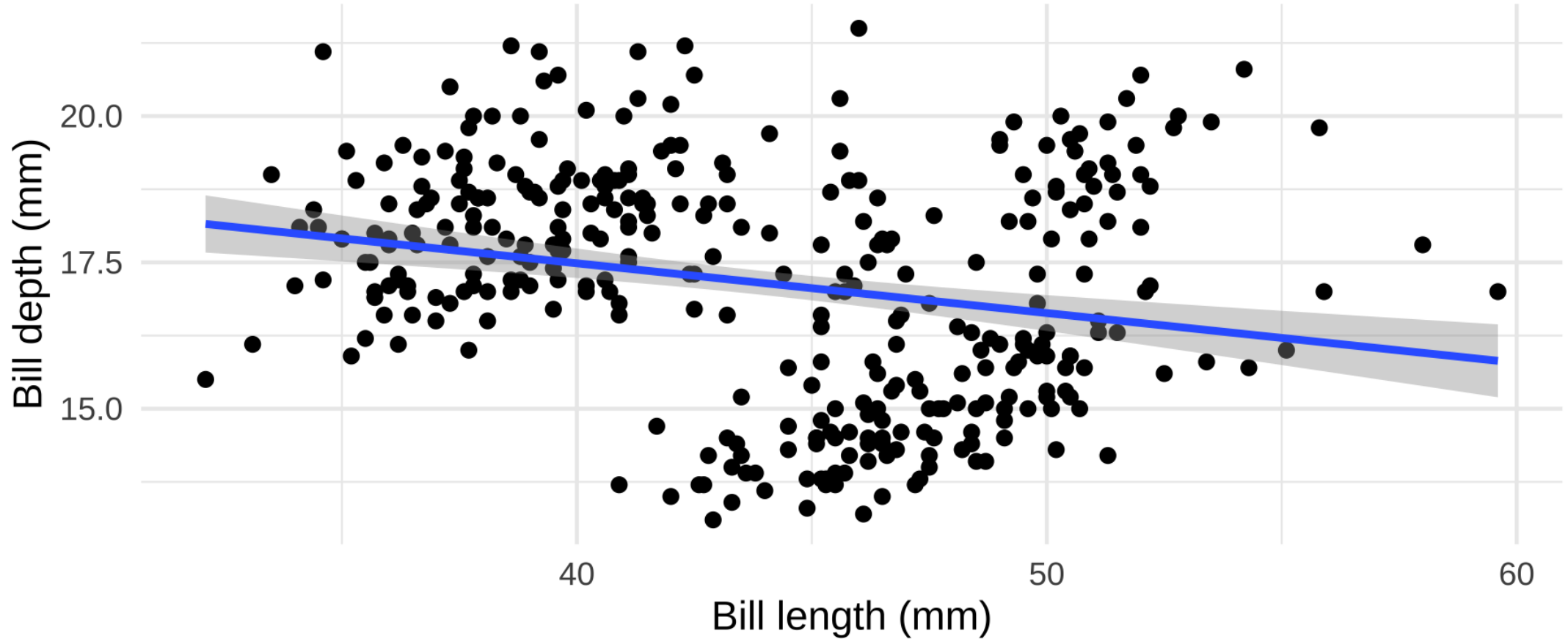
Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

```
library(palmerpenguins)
```

How does bill length relate to bill depth?

Penguin bill dimensions

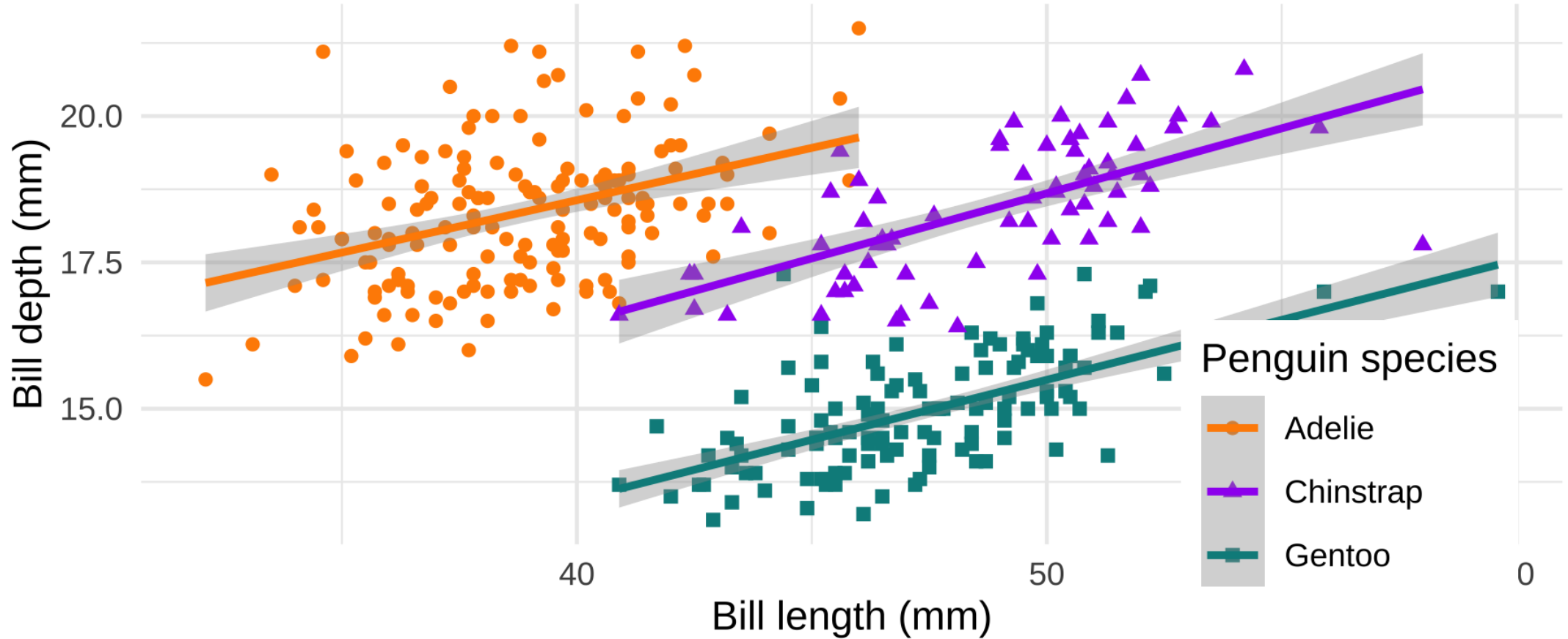
Palmer Station LTER



How does bill length relate to bill depth?

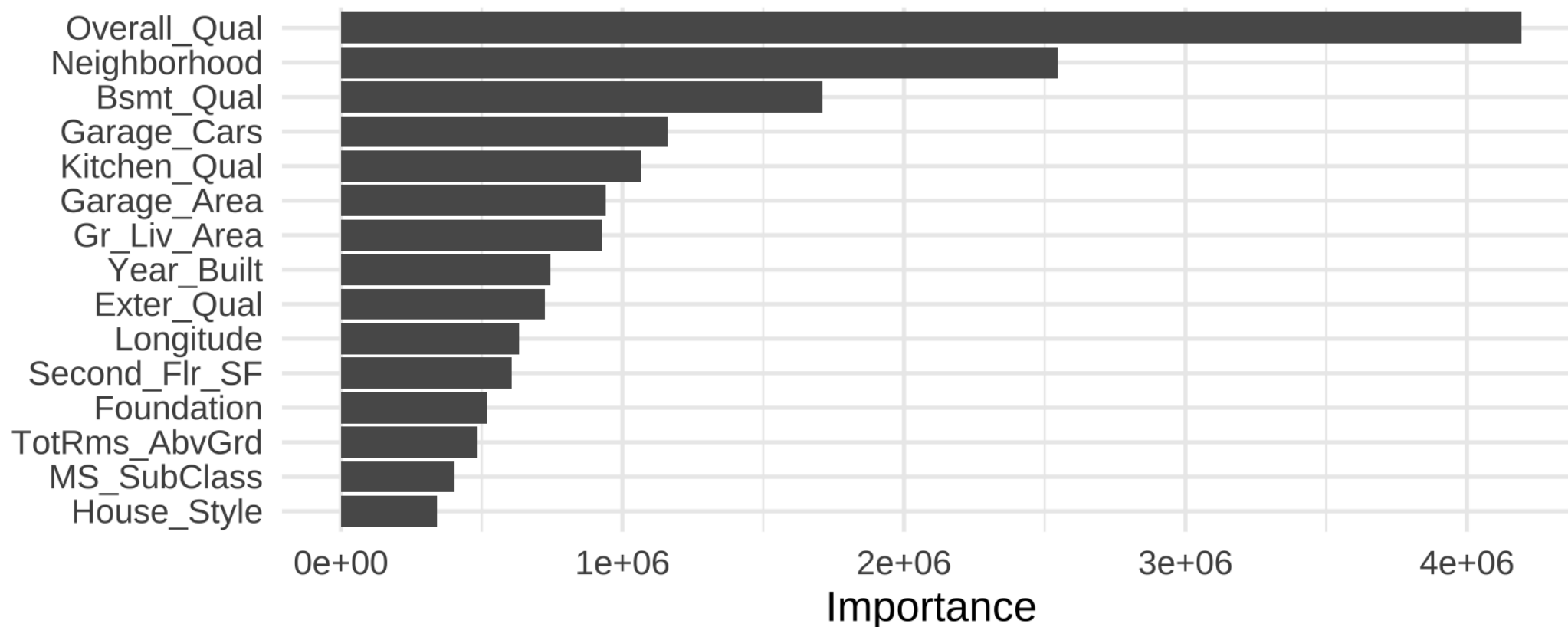
Penguin bill dimensions

Bill length and depth for Adelie, Chinstrap and Gentoo Penguins at Palmer Station



Variable Importance Plots

```
regresion_workflow <- workflow() %>% add_model(decision_tree(mode = "regression") %>% set_engine('rpart'))  
model <- regresion_workflow %>%  
  add_recipe(recipe(Sale_Price ~ ., data = ames_train)) %>%  
  fit(data = ames_train)  
model %>% pull_workflow_fit() %>% vip::vip(num_features = 15L)
```



How much does it help to have a feature in?

```
set.seed(20201118)
resamples <- vfold_cv(ames_train, v = 10)
```

```
regresion_workflow %>%
  add_recipe(recipe(Sale_Price ~ ., data = ames_train)) %>%
  fit_resamples(resamples = resamples, metrics = metric_set(mae, rmse)) %>% collect_metrics()
```

.metric	.estimator	mean	n	std_err
<chr>	<chr>	<dbl>	<int>	<dbl>
mae	standard	25.53709	10	0.5663083
rmse	standard	36.71345	10	0.9225024

2 rows

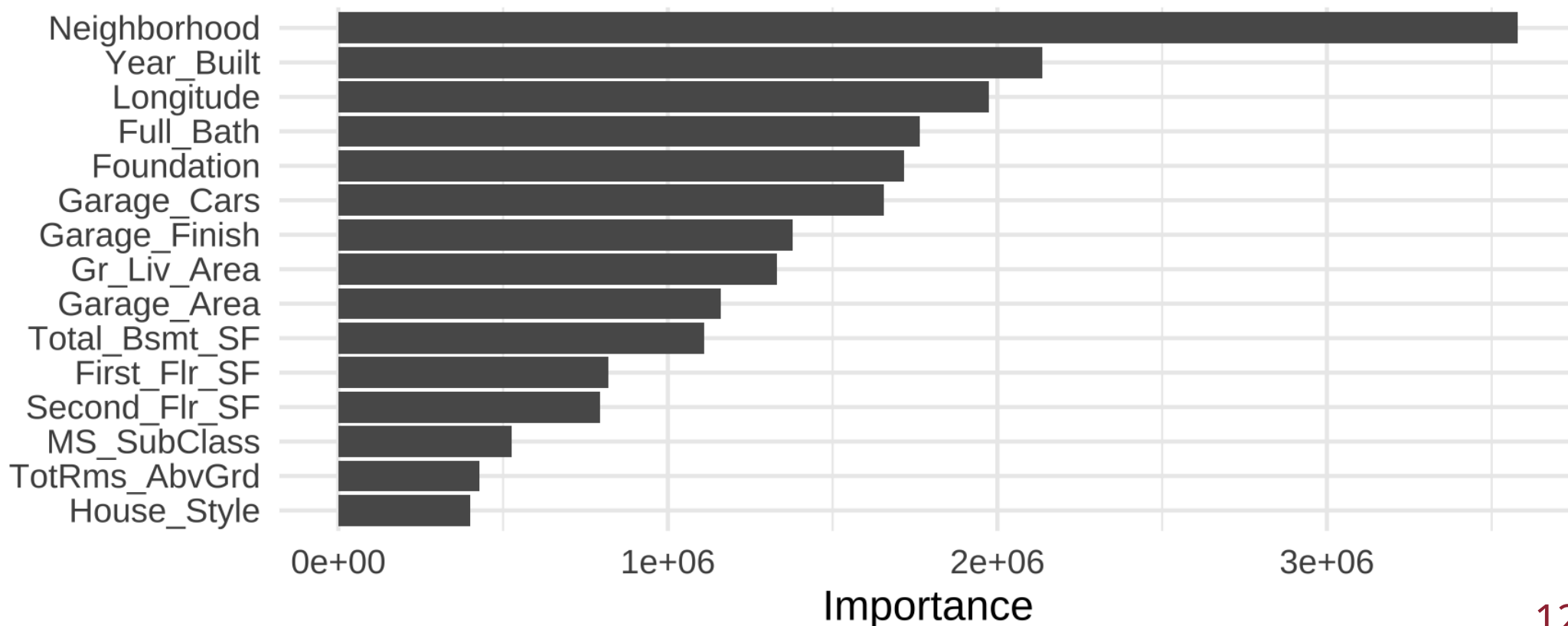
```
regresion_workflow %>%
  add_recipe(recipe(Sale_Price ~ ., data = ames_train) %>% step_rm(ends_with("Qual"))) %>%
  fit_resamples(resamples = resamples, metrics = metric_set(mae, rmse)) %>% collect_metrics()
```

.metric	.estimator	mean	n	std_err
<chr>	<chr>	<dbl>	<int>	<dbl>
mae	standard	25.36555	10	0.6889984
rmse	standard	36.45790	10	0.9566548

2 rows

Variable Importance without the Quality Features

```
regression_workflow %>%  
  add_recipe(recipe(Sale_Price ~ ., data = ames_train) %>% step_rm(ends_with("Qual")) %>%  
  fit(data = ames_train) %>%  
  pull_workflow_fit() %>% vip::vip(num_features = 15L)
```



Appendix: code

```
include_graphics("https://raw.githubusercontent.com/allisonhorst/palmerpenguins/master/man/figures/logo.png")
include_graphics("https://raw.githubusercontent.com/allisonhorst/palmerpenguins/master/man/figures/culmen_depth.png")
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Penguin bill dimensions", subtitle = "Palmer Station LTER", x = "Bill length (mm)", y = "Bill depth (mm)")
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species, shape = species)) +
  geom_point() +
  geom_smooth(method = "lm") +
  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +
  labs(title = "Penguin bill dimensions",
       subtitle = "Bill length and depth for Adelie, Chinstrap and Gentoo Penguins at Palmer Station LTER",
       x = "Bill length (mm)",
       y = "Bill depth (mm)",
       color = "Penguin species",
       shape = "Penguin species") +
  theme(legend.position = c(0.85, 0.15),
        legend.background = element_rect(fill = "white", color = NA))
#data(ames, package = "modeldata")
ames <- AmesHousing::make_ames()
ames_all <- ames %>%
  filter(Gr_Liv_Area < 4000, Sale_Condition == "Normal") %>%
  mutate(across(where(is.integer), as.double)) %>%
  mutate(Sale_Price = Sale_Price / 1000)
rm(ames)
set.seed(10) # Seed the random number generator
ames_split <- initial_split(ames_all, prop = 2 / 3)
ames_train <- training(ames_split)
ames_test <- testing(ames_split)
```